

EquiRisk: Operationalizing Equity-Aware Risk Stratification for Diabetes Management in Food Deserts

Min Wu, PhD¹

1. Zilber College of Public Health, University of Wisconsin-Milwaukee, US

Abstract

Traditional diabetes risk stratification often focuses on clinical indicators without integrating contextual factors, limiting actionable interventions in high-need settings such as food deserts. We introduce EquiRisk, a conceptual framework designed to integrate individual clinical risk, neighborhood-level vulnerability, and action feasibility to enable operational, equity-aware public health decision-making. Using diabetes management in food deserts as an illustrative case, EquiRisk shifts equity from a retrospective reporting metric into a proactive decision rule. The framework supports dynamic prioritization, resource alignment, and scenario-aware intervention planning. EquiRisk demonstrates how AI-enabled tools can provide bounded, policy-constrained coordination to translate risk signals into actionable pathways, preserving human oversight and accountability. The approach has implications for equitable program design, workforce training, and scalable public health interventions.

Keywords: *Equity, Risk Stratification, Diabetes, Food Deserts, Public Health Informatics, AI, Decision Support*

Introduction

Diabetes risk stratification is central to population health management. However, in food deserts, traditional approaches may fail to account for structural barriers, including limited access to healthy food and transportation, creating a disconnect between risk identification and feasible intervention. Public health systems generate relevant data (e.g., clinical risk scores¹, Social Vulnerability Index², Area Deprivation Index^{3,4}, program inventories), yet these elements remain fragmented and underutilized in operational decision-making. EquiRisk addresses this gap by integrating existing data into an actionable, equity-aware framework. The conceptual framework described here (EquiRisk) will be detailed in an upcoming Springer textbook on AI in public health⁵, expected March 2026.

Framework Description

EquiRisk combines three interrelated layers (see Figure 1):

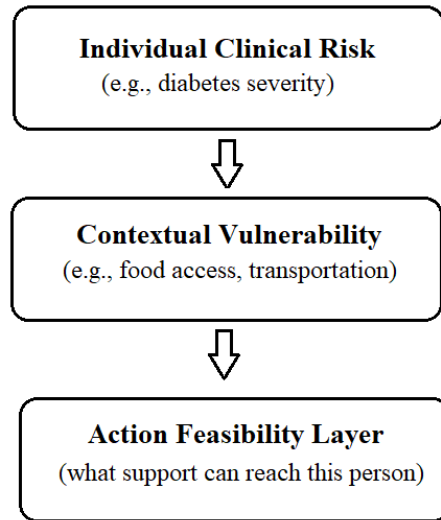


Figure 1. The EquiRisk Framework: Integrating Risk, Context, and Action

- 1) **Individual Clinical Risk:** Identifies individuals with elevated health risk, providing the foundation for prioritization.
- 2) **Contextual Vulnerability:** Adjusts prioritization based on environmental and social determinants, including food access, transportation, housing stability, and neighborhood resource availability.
- 3) **Action Feasibility:** Evaluates the practicality of interventions given current programs, staffing, and policy constraints, translating risk and context into feasible, operationally relevant outputs.

Implementation and Operational Logic

EquiRisk can be applied manually or supported by digital tools, including AI-enabled systems that coordinate disparate data streams without autonomous decision-making. The framework prioritizes patients based on combined clinical risk, contextual vulnerability, and intervention feasibility, allowing public health teams to answer: "Who can we help now, and how?" This approach facilitates scenario-aware prioritization, resource alignment, and improved intervention effectiveness without additional budget requirements.

Illustrative Case: Diabetes in Food Deserts

Two hypothetical patients with similar clinical risk illustrate EquiRisk's operational logic (see Table 1).

Dimension	Patient A (Not in Food Desert)	Patient B (In Food Desert)
Baseline Clinical Risk and Likelihood of Uptake	<i>Traditional:</i> High clinical risk based on HbA1c, age, and comorbidities; high likelihood of uptake due to access and stability. <i>EquiRisk:</i> Clinical risk confirmed; no contextual elevation needed.	<i>Traditional:</i> High clinical risk based on clinical indicators; likelihood of uptake not assessed. <i>EquiRisk:</i> Clinical risk elevated by low likelihood of uptake due to access, mobility, and resource constraints.
Neighborhood Context	<i>Traditional:</i> Neighborhood context not incorporated into risk scoring. <i>EquiRisk:</i> Adequate food access, transportation, and service density confirm feasibility of standard care.	<i>Traditional:</i> Neighborhood context not incorporated. <i>EquiRisk:</i> Food desert status, limited transportation, and low service density formally increase contextual vulnerability.
Feasibility of Standard Recommendations	<i>Traditional:</i> Nutrition counseling and clinic follow-up assumed feasible. <i>EquiRisk:</i> Standard recommendations remain appropriate and feasible.	<i>Traditional:</i> Standard recommendations issued despite limited feasibility. <i>EquiRisk:</i> Feasibility constraints trigger alternative, supportive interventions.
Decision Logic and Contextual Adjustment	<i>Traditional:</i> Prioritization driven primarily by clinical risk. <i>EquiRisk:</i> Clinical risk sufficient to guide routine prioritization.	<i>Traditional:</i> Same prioritization logic as Patient A. <i>EquiRisk:</i> Clinical risk combined with contextual vulnerability drives higher priority and tailored response.
Resource Matching Quality and Efficiency	<i>Traditional:</i> Standard outpatient services align with available resources. <i>EquiRisk:</i> High resource-intervention match maintained.	<i>Traditional:</i> Outreach efforts are likely mismatched and inefficient. <i>EquiRisk:</i> Resources redirected toward food support, transportation, and community health worker engagement.
Expected Public Health Impact	<i>Traditional:</i> Improved glycemic control expected. <i>EquiRisk:</i> Similar outcomes achieved with efficient resource use.	<i>Traditional:</i> Limited improvement due to low uptake and follow-through. <i>EquiRisk:</i> Improved glycemic control, reduced emergency visits, and more equitable outcomes.

Table 1. Contrasting Traditional and Equity-Aware Risk Stratification for Two Patients With Similar Clinical Risk

While traditional models prioritize solely on clinical risk, EquiRisk adjusts for contextual barriers and recommends feasible interventions (e.g., alternative nutrition support, transportation

assistance, or community health worker engagement). This approach enhances equity, efficiency, and actionable decision-making.

Safety, Trust, and Governance

EquiRisk is a domain-specific, decision-support framework rather than general AI. It preserves human oversight, professional judgment, and accountability. Design principles include transparency, auditable decision pathways, and policy-constrained automation, ensuring ethical and trustworthy application in public health practice.

Implications

Conceptually, EquiRisk operationalizes equity as a decision rule. Methodologically, it enables dynamic, scenario-aware risk stratification. Practically, it enhances resource allocation efficiency. Educationally, it provides a teachable model for public health training in equity-aware decision-making.

Conclusion

EquiRisk demonstrates how integrating clinical risk, social context, and action feasibility can improve equity, efficiency, and operational relevance in public health interventions. As AI-enabled coordination tools become increasingly feasible, frameworks like EquiRisk provide practical guidance for translating data into actionable, ethically grounded public health action.

References

1. Kautter J, Pope GC, Ingber M, et al. The HHS-HCC risk adjustment model for individual and small group markets under the Affordable Care Act. *Medicare & Medicaid Research Review*. 2014;4(3):mmrr2014-2004-2003-a2003.
2. Centers for Disease C, Prevention, Agency for Toxic S, Disease R. CDC/ATSDR Social Vulnerability Index. 2026; <https://www.atsdr.cdc.gov/place-health/php/svi/index.html>. Accessed January 8, 2026, 2026.
3. Kind AJH, Buckingham W. Making neighborhood-disadvantage metrics accessible: The Neighborhood Atlas. *New England Journal of Medicine*. 2018;378:2456-2458.
4. Wisconsin-Madison Uo. Neighborhood Atlas - Area Deprivation Index. 2026; <https://www.neighborhoodatlas.medicine.wisc.edu/>.
5. Wu M. *Artificial Intelligence in Public Health*. New York, NY, USA: Springer Nature; 2026.